TEMA: L3. Analytical Integration Model: Implementare OLAP views

Sisteme Informaționale Integrate
Model analitic federativ – Jocurile Olimpice

Indicativ echipă: **SIA04**

Implementare OLAP views și model analitic federativ

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1 Nivel CONSOLIDARE date

1.1 View_Consolidare_1: INT_RESULTS_FULL_V

Descriere: Integrare completă a rezultatelor olimpice, consolidând date din PostgreSQL (rezultate, atleți) și Oracle (ediții, evenimente, sporturi, țări).

Surse de date integrate: DS_1 (Oracle/OLY_REF), DS_2 (PostgreSQL/PG_RESULTS, PG_ATHLETES)

Definiție DDL SQL:

```sql
CREATE OR REPLACE VIEW FDBO.INT_RESULTS_FULL_V AS
SELECT
    r.result_id,
    r.athlete_id,
    a.full_name        AS athlete_name,
    a.sex,
    r.age,
    a.height_cm,
    a.weight_kg,
    r.team,
    r.medal,
    g.game_id,
    g.games_name,
    g.year             AS game_year,
    g.season,
    g.city,
    e.event_id,
    e.event_name,
    s.sport_id,
    s.sport_name,
    c.noc,
    c.region            AS country_name,
    c.notes             AS country_notes,
    'PG+Oracle' AS source_type
FROM FDBO.PG_RESULTS r
JOIN FDBO.PG_ATHLETES a   ON r.athlete_id = a.athlete_id
LEFT JOIN OLY_REF.GAMES g     ON r.game_id    = g.game_id
LEFT JOIN OLY_REF.EVENTS e    ON r.event_id   = e.event_id
LEFT JOIN OLY_REF.SPORTS s    ON e.sport_id   = s.sport_id
LEFT JOIN OLY_REF.COUNTRIES c ON r.noc         = c.noc;
```

2 Schema analitică ROLAP

2.1 Tabele de fapte (ROLAP Fact Views)

2.1.1 Tabela_de_fapte_1: FACT_RESULTS_V

Descriere: Faptă centrală – participarea unui atlet la un eveniment, cu detalii despre medalie, vârstă, echipă, ediție.

Surse de date integrate: DS_1 (Oracle/OLY_REF), DS_2 (PostgreSQL/PG_RESULTS)

Definiție DDL SQL:

```sql
CREATE OR REPLACE VIEW FDBO.FACT_RESULTS_V AS
SELECT
    r.result_id,
    r.athlete_id,
    r.game_id,
    r.event_id,
    r.noc,
    r.medal,
    r.age,
    r.team,
    CASE WHEN r.medal = 'Gold'   THEN 1 ELSE 0 END AS is_gold,
    CASE WHEN r.medal = 'Silver' THEN 1 ELSE 0 END AS is_silver,
    CASE WHEN r.medal = 'Bronze' THEN 1 ELSE 0 END AS is_bronze,
    CASE WHEN r.medal IS NOT NULL THEN 1 ELSE 0 END AS has_medal,
    g.year AS game_year,
    g.season
FROM FDBO.PG_RESULTS r
LEFT JOIN OLY_REF.GAMES g ON r.game_id = g.game_id;
```

2.2 Tabele dimensionale (OLAP Dimensional Views)

2.2.1 Tabela_dimensionala_1: DIM_ATHLETE_V

Descriere: Dimensiune atleți – date personale, țară.

Surse de date integrate: DS_2 (PostgreSQL/PG_ATHLETES), DS_1 (Oracle/-COUNTRIES)

Definiție DDL SQL:

```sql
CREATE OR REPLACE VIEW FDBO.DIM_ATHLETE_V AS
SELECT DISTINCT
    a.athlete_id,
    a.full_name,
    a.sex,
    a.birth_year,
    a.height_cm,
    a.weight_kg,
    a.noc,
    c.region AS country_name
FROM FDBO.PG_ATHLETES a
LEFT JOIN OLY_REF.COUNTRIES c ON a.noc = c.noc;
```

2.2.2 Tabela_dimensionala_2: DIM_GAME_V

Descriere: Dimensiune ediții olimpice – an, sezon, oraș, eră.

Surse de date integrate: DS_1 (Oracle/GAMES)

Definiție DDL SQL:

```sql
CREATE OR REPLACE VIEW FDBO.DIM_GAME_V AS
SELECT
```

```
    g.game_id,
    g.games_name,
    g.year,
    g.season,
    g.city,
    CASE
        WHEN g.year < 1920 THEN 'Pionier (pre-1920)'
        WHEN g.year < 1950 THEN 'Interbelic (1920-1948)'
        WHEN g.year < 1980 THEN 'Epoca de Aur (1952-1976)'
        WHEN g.year < 2000 THEN 'Modern (1980-1996)'
        ELSE 'Contemporan (2000+)'
    END AS era
FROM OLY_REF.GAMES g;
```

2.2.3 Tabela_dimensionala_3: DIM_EVENT_V

Descriere: Dimensiune evenimente – sport, categorie de gen.

Surse de date integrate: DS_1 (Oracle/EVENTS, SPORTS)

Definiție DDL SQL:

```
CREATE OR REPLACE VIEW FDBO.DIM_EVENT_V AS
SELECT
    e.event_id,
    e.event_name,
    s.sport_id,
    s.sport_name,
    CASE
        WHEN INSTR(e.event_name, 'Men''s') > 0 THEN 'Men'
        WHEN INSTR(e.event_name, 'Women''s') > 0 THEN 'Women'
        ELSE 'Mixed'
    END AS gender_category
FROM OLY_REF.EVENTS e
JOIN OLY_REF.SPORTS s ON e.sport_id = s.sport_id;
```

3 View-uri OLAP analitice (agregări, ROLLUP, CUBE, GROUPING SETS)

3.1 View_Analitic_OLAP_1: OLAP_ROLLUP_COUNTRY_SPORT_V

Tip procesare: ROLLUP (țară, sport) – subtotaluri pe țară și sport.

```
CREATE OR REPLACE VIEW FDBO.OLAP_ROLLUP_COUNTRY_SPORT_V AS
SELECT
    c.region                                    AS country_name,
    s.sport_name,
    COUNT(*)                                    AS participations,
    SUM(CASE WHEN r.medal IS NOT NULL THEN 1 ELSE 0 END) AS
       total_medals,
    SUM(CASE WHEN r.medal = 'Gold'   THEN 1 ELSE 0 END)  AS gold,
    SUM(CASE WHEN r.medal = 'Silver' THEN 1 ELSE 0 END)  AS silver,
```

```sql
    SUM(CASE WHEN r.medal = 'Bronze' THEN 1 ELSE 0 END)  AS bronze,
    GROUPING(c.region)                                   AS is_country_total,
    GROUPING(s.sport_name)                               AS is_sport_total
FROM FDBO.PG_RESULTS r
JOIN OLY_REF.COUNTRIES c ON r.noc = c.noc
JOIN OLY_REF.EVENTS e    ON r.event_id = e.event_id
JOIN OLY_REF.SPORTS s    ON e.sport_id = s.sport_id
WHERE r.medal IS NOT NULL
GROUP BY ROLLUP(c.region, s.sport_name)
ORDER BY c.region NULLS LAST, s.sport_name NULLS LAST;
```

3.2 View_Analitic_OLAP_2: OLAP_CUBE_COUNTRY_SEASON_V

Tip procesare: CUBE (țară, sezon) – totaluri pe toate combinațiile.

```sql
CREATE OR REPLACE VIEW FDBO.OLAP_CUBE_COUNTRY_SEASON_V AS
SELECT
    c.region                                             AS country_name,
    g.season,
    COUNT(DISTINCT r.athlete_id)                         AS unique_athletes,
    SUM(CASE WHEN r.medal IS NOT NULL THEN 1 ELSE 0 END) AS
       total_medals,
    SUM(CASE WHEN r.medal = 'Gold'   THEN 1 ELSE 0 END)  AS gold,
    SUM(CASE WHEN r.medal = 'Silver' THEN 1 ELSE 0 END)  AS silver,
    SUM(CASE WHEN r.medal = 'Bronze' THEN 1 ELSE 0 END)  AS bronze,
    GROUPING(c.region)                                   AS is_country_total,
    GROUPING(g.season)                                   AS is_season_total
FROM FDBO.PG_RESULTS r
JOIN OLY_REF.COUNTRIES c ON r.noc = c.noc
JOIN OLY_REF.GAMES g     ON r.game_id = g.game_id
WHERE r.medal IS NOT NULL
GROUP BY CUBE(c.region, g.season)
ORDER BY c.region NULLS LAST, g.season NULLS LAST;
```

3.3 View_Analitic_OLAP_3: OLAP_GS_DATA_SOURCES_V

Tip procesare: GROUPING SETS – statistici pe surse de date și categorii.

```sql
CREATE OR REPLACE VIEW FDBO.OLAP_GS_DATA_SOURCES_V AS
SELECT
    source_name,
    category,
    SUM(record_count) AS record_count,
    GROUPING(source_name) AS is_source_total,
    GROUPING(category)    AS is_category_total,
    GROUPING_ID(source_name, category) AS grouping_id
FROM (
    SELECT 'DS1-Oracle' AS source_name, 'Sports'    AS category,
       COUNT(*) AS record_count FROM OLY_REF.SPORTS
    UNION ALL
    SELECT 'DS1-Oracle', 'Events',    COUNT(*) FROM OLY_REF.EVENTS
```

```sql
    UNION ALL
    SELECT 'DS1-Oracle', 'Games',     COUNT(*) FROM OLY_REF.GAMES
    UNION ALL
    SELECT 'DS1-Oracle', 'Countries', COUNT(*) FROM OLY_REF.COUNTRIES
    UNION ALL
    SELECT 'DS2-PostgreSQL', 'Athletes', COUNT(*) FROM FDBO.PG_ATHLETES
    UNION ALL
    SELECT 'DS2-PostgreSQL', 'Results',  COUNT(*) FROM FDBO.PG_RESULTS
    UNION ALL
    SELECT 'DS3-JSON/MongoDB', 'Medal Docs',   COUNT(*) FROM FDBO.JSON_MEDAL_DOCS WHERE doc_type = 'athlete_medal'
    UNION ALL
    SELECT 'DS3-JSON/MongoDB', 'Game Summaries', COUNT(*) FROM FDBO.JSON_MEDAL_DOCS WHERE doc_type = 'game_summary'
    UNION ALL
    SELECT 'DS4-CSV', 'Enriched Events', COUNT(*) FROM FDBO.CSV_ATHLETE_EVENTS
) src
GROUP BY GROUPING SETS (
    (source_name, category),
    (source_name),
    ()
)
ORDER BY grouping_id, source_name NULLS LAST, category NULLS LAST;
```

4 Resurse REST și pagini Web

4.1 REST API (ORDS)

9 endpoint-uri publicate (ex: /olympics/sports, /olympics/games, /olympics/top-medalists, /olympics/olap/country-medals etc.)
Script: CLOUD_10_ORDS_REST.sql

4.2 Pagini APEX

Orice view de mai sus poate fi sursă pentru regiuni de tip REPORT sau CHART (ex: OLAP_ROLLUP_COUNTRY_SPORT_V pentru chart medalii pe țări/sporturi).

5 Alte fișiere relevante

- CLOUD_07_FDBO_OLAP_VIEWS.sql (dacă există)
- CLOUD_10_ORDS_REST.sql (REST API)
- create_views.py (automatizare creare view-uri)

- README.md (documentație completă)
- Toate scripturile SQL din Proiect/2. Access Model/ și Proiect/3. Integration Model/